

Questions

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Q1 - 2024 (04 Apr Shift 1)

A metal wire of uniform mass density having length L and mass M is bent to form a semicircular arc and a particle of mass m is placed at the centre of the arc. The gravitational force on the particle by the wire is :

(1) $\frac{GmM\pi^2}{L^2}$

(2) $\frac{GMm\pi}{2L^2}$

(3) 0

(4) $\frac{2GmM\pi}{L^2}$

Q2 - 2024 (04 Apr Shift 2)

A 90 kg body placed at $2R$ distance from surface of earth experiences gravitational pull of :

(R = Radius of earth, $g = 10 \text{ m s}^{-2}$)

(1) 100 N

(2) 300 N

(3) 225 N

(4) 120 N

Q3 - 2024 (04 Apr Shift 2)

Correct formula for height of a satellite from earth's surface is:

(1) $\left(\frac{T^2 R^2}{4\pi^2 g}\right)^{1/3} - R$

(2) $\left(\frac{T^2 R^2 g}{4\pi^2}\right)^{1/3} - R$

(3) $\left(\frac{T^2 R^2 g}{4\pi^2}\right)^{-1/3} + R$

(4) $\left(\frac{T^2 R^2 g}{4\pi}\right)^{1/2} - R$

Q4 - 2024 (05 Apr Shift 1)

A simple pendulum doing small oscillations at a place R height above earth surface has time period of

$T_1 = 4 \text{ s}$. T_2 would be its time period if it is brought to a point which is at a height $2R$ from earth surface.

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Choose the correct relation [$R = \text{radius of earth}$] :

(1) $2 T_1 = T_2$

(2) $2 T_1 = 3 T_2$

(3) $T_1 = T_2$

(4) $3 T_1 = 2 T_2$

Q5 - 2024 (05 Apr Shift 1)

In hydrogen like system the ratio of coulombian force and gravitational force between an electron and a proton is in the order of :

(1) 10^{39}

(2) 10^{29}

(3) 10^{19}

(4) 10^{36}

Q6 - 2024 (05 Apr Shift 1)

Match List I with List II :

List I

List II

(A) Kinetic energy of planet

(I) $-\frac{GMm}{a}$

(B) Gravitation Potential energy of sun-planet system

(II) $\frac{GMm}{2a}$

(C) Total mechanical energy of planet

(III) $\frac{Gm}{r}$

(D) Escape energy at the surface of planet for unit mass object

(IV) $-\frac{GMm}{2a}$

(Where $a = \text{radius of planet orbit}$, $r = \text{radius of planet}$, $M = \text{mass of Sun}$, $m = \text{mass of planet}$)

Choose the correct answer from the options given below :

(1) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(4) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

Q7 - 2024 (05 Apr Shift 2)

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A satellite revolving around a planet in stationary orbit has time period 6 hours. The mass of planet is one-fourth the mass of earth. The radius orbit of planet is :

(Given = Radius of geo-stationary orbit for earth is 4.2×10^4 km)

(1) 1.4×10^4 km

(2) 1.05×10^4 km

(3) 8.4×10^4 km

(4) 1.68×10^5 km

Q8 - 2024 (06 Apr Shift 1)

To project a body of mass m from earth's surface to infinity, the required kinetic energy is (assume, the radius of earth is R_E , g = acceleration due to gravity on the surface of earth):

(1) $2mgR_E$

(2) $4mgR_E$

(3) mgR_E

(4) $1/2mgR_E$

Q9 - 2024 (06 Apr Shift 2)

Assuming the earth to be a sphere of uniform mass density, a body weighed 300 N on the surface of earth.

How much it would weigh at $R/4$ depth under surface of earth ?

(1) 75 N

(2) 300 N

(3) 375 N

(4) 225 N

Q10 - 2024 (08 Apr Shift 1)

Two planets A and B having masses m_1 and m_2 move around the sun in circular orbits of r_1 and r_2 radii respectively. If angular momentum of A is L and that of B is $3L$, the ratio of time period $\left(\frac{T_A}{T_B}\right)$ is:

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- (1) $\left(\frac{r_2}{r_1}\right)^{\frac{3}{2}}$
- (2) $\frac{1}{27}\left(\frac{m_2}{m_1}\right)^3$
- (3) $27\left(\frac{m_1}{m_2}\right)^3$
- (4) $\left(\frac{r_1}{r_2}\right)^3$

Q11 - 2024 (08 Apr Shift 2)

Two satellite A and B go round a planet in circular orbits having radii $4R$ and R respectively. If the speed of A is $3v$, the speed of B will be :

- (1) $3v$
- (2) $6v$
- (3) $\frac{4}{3}v$
- (4) $12v$

Q12 - 2024 (09 Apr Shift 1)

An astronaut takes a ball of mass m from earth to space. He throws the ball into a circular orbit about earth at an altitude of 318.5 km. From earth's surface to the orbit, the change in total mechanical energy of the ball is $x \frac{GM_e m}{21R_e}$. The value of x is (take $R_e = 6370$ km) :

- (1) 10
- (2) 12
- (3) 9
- (4) 11

Q13 - 2024 (09 Apr Shift 2)

A satellite of 10^3 kg mass is revolving in circular orbit of radius $2R$. If $\frac{10^4 R}{6} J$ energy is supplied to the satellite, it would revolve in a new circular orbit of radius

(use $g = 10 \text{ m/s}^2$, $R = \text{radius of earth}$)

- (1) $2.5R$

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Questions

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Answer Key

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Q1 (4) mathongo Q2 (1) mathongo Q3 (2) mathongo Q4 (4) mathongo

Q5 (1) mathongo Q6 (2) mathongo Q7 (2) mathongo Q8 (3) mathongo

Q9 (4) mathongo Q10 (2) mathongo Q11 (2) mathongo Q12 (4) mathongo

Q13 (4) mathongo mathongo mathongo mathongo mathongo mathongo

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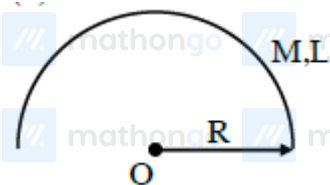
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Solutions

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Q1



We have $R = \frac{L}{\pi}$

$$g_0 = \frac{2G \frac{M}{L}}{R} = \frac{2GM\pi}{L^2}$$

$$\therefore F_m = mg_0 = \frac{2GM\pi m}{L^2}$$

Hence option (4)

Q2

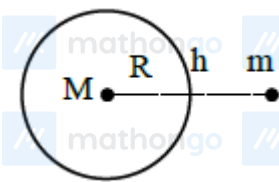
Here g_s = gravitational acceleration at surface

$$\text{Value of } g = g_s \left(1 + \frac{h}{R}\right)^{-2}$$

$$= g_s (1 + 2)^{-2} = \frac{g_s}{9}$$

$$\text{Force} = mg = 90 \times \frac{g_s}{9} = 100 \text{ N}$$

Q3



$$\Rightarrow \frac{GMm}{(R+h)^2} = \frac{mv^2}{(R+h)}$$

$$\Rightarrow \frac{GM}{(R+h)} = v^2 \dots (1)$$

$$\Rightarrow v = (R+h)\omega$$

$$\Rightarrow v = (R+h) \frac{2\pi}{T} \dots (2)$$

$$\Rightarrow \frac{GM}{R^2} = g$$

$$\Rightarrow GM = gR^2 \dots (3)$$

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Solutions

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Put value from (2) & (3) in eq. (1)

$$\Rightarrow \frac{gR^2}{(R+h)} = (R+h)^2 \left(\frac{2\pi}{T} \right)^2$$

$$\Rightarrow \frac{T^2 R^2 g}{(2\pi)^2} = (R+h)^3$$

$$\Rightarrow \left[\frac{T^2 R^2 g}{(2\pi)^2} \right]^{1/3} - R = h$$

Q4

$$T_1 = 2\pi \sqrt{\frac{\ell}{GM}} (2R)^2$$

$$T_2 = 2\pi \sqrt{\frac{\ell}{GM}} (3R)^2$$

$$\therefore \frac{T_1}{T_2} = \frac{2}{3}$$

Q5

$$F_e = \frac{kQ_1 Q_2}{r^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{r^2}$$

$$F_g = \frac{Gm_1 m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-27}}{r^2}$$

$$\frac{F_e}{F_g} \cong 0.23 \times 10^{40} \cong 2.3 \times 10^{39}$$

Option (1)

Q6

$$KE = \frac{1}{2}mv^2 = \frac{GMm}{2a}$$

$$PE = -2KE$$

$$TE = -KE$$

Q7

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Solutions

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$$T = \frac{2\pi r^{3/2}}{\sqrt{GM}}$$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2} \left(\frac{M_2}{M_1}\right)^{1/2}$$

$$\frac{6}{24} = \frac{(r_1)^{3/2}}{(4.2 \times 10^4)^{3/2}} \left(\frac{M}{M/4}\right)^{1/2}$$

$$r_1 = 1.05 \times 10^4 \text{ km}$$

Q8

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R_E}$$

$$g = \frac{GM}{R_E^2}$$

$$K = mgR_E$$

Q9

At surface: $mg = 300 \text{ N}$

$$m = \frac{300}{g_s}$$

$$\text{At Depth } \frac{R}{4} : g_d = g_s \left[1 - \frac{d}{R}\right]$$

$$g_d = g_s \left[1 - \frac{R}{4R}\right]$$

$$g_d = \frac{3}{4} g_s$$

$$\text{weight at depth} = m \times g_d$$

$$= m \times \frac{3}{4} g_s$$

$$= \frac{3}{4} \times 300$$

$$= 225 \text{ N}$$

Q10

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Solutions

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$$\frac{\pi r_1^2}{T_A} = \frac{L}{2m_1} \dots\dots\dots (1)$$

$$\frac{\pi r_2^2}{T_B} = \frac{3L}{2m_2} \dots\dots\dots (2)$$

$$\Rightarrow \frac{T_A}{T_B} = 3 \cdot \frac{m_1}{m_2} \cdot \left(\frac{r_1}{r_2}\right)^2$$

$$\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_1}{r_2}\right)^3 \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{T_A}{T_B}\right)^{\frac{4}{3}}$$

$$\Rightarrow \frac{1}{27} \cdot \left(\frac{m_2}{m_1}\right)^3 = \left(\frac{T_A}{T_B}\right)$$

Q11

$$v = \sqrt{\frac{GM}{R}}$$

$$\frac{v_A}{v_B} = \sqrt{\frac{R_B}{R_A}} = \sqrt{\frac{R}{4R}} = \frac{1}{2}$$

$$v_B = 2v_A = 6v$$

Q12

$$h = 318.5 \approx \left(\frac{R_e}{20}\right)$$

$$T \cdot E_i = \frac{-GM_e m}{R_e}$$

$$T \cdot E_f = \frac{-GM_e m}{2(R_e + h)} = \frac{-GM_e m}{2\left(R_e + \frac{R_e}{20}\right)}$$

$$\Rightarrow T \cdot E_f = \frac{-10GM_e m}{21R_e}$$

Change in total mechanical energy

$$= TE_f - TE_i$$

$$= \frac{GM_e m}{Re} \left[1 - \frac{10}{21}\right] = \frac{11GM_e m}{21 Re}$$

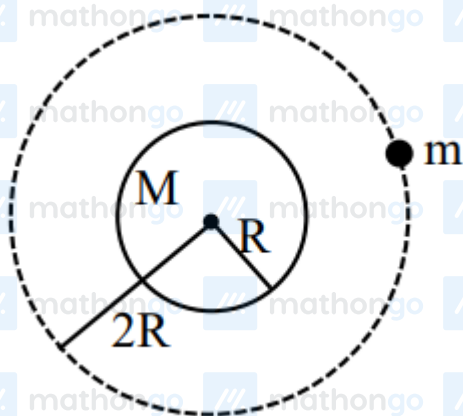
Q13

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Solutions

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$$\text{Total energy} = \frac{-GMm}{2(2R)}$$

if energy $= \frac{10^4 R}{6}$ is added then

$$\frac{-GMm}{4R} + \frac{10^4 R}{6} = \frac{-GMm}{2r}$$

where r is new radius of revolving and $g = \frac{GM}{R^2}$

$$-\frac{mgR}{4} + \frac{10^4 R}{6} = -\frac{mgR^2}{2r} \quad (m = 10^3 \text{ kg})$$

$$-\frac{10^3 \times 10 \times R}{4} + \frac{10^4 R}{6} = -\frac{10^3 \times 10 \times R^2}{2r}$$

$$-\frac{1}{4} + \frac{1}{6} = -\frac{R}{2r}$$

$$r = 6R$$

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